

Trends in Renewable Energy Generation

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This study examines the trends in renewable energy generation, both in absolute terms and as a percentage of the PES (Primary Energy Supply), across three regions: *East Asia & Pacific*, *Europe & Central Asia* and *Latin America & Caribbean*. By visualising 20 years of data in Python, we observe a steady increase in total renewable energy generation, but less consistent changes in relation to the total energy supply. This study also finds a negative correlation between GDP and the contribution of fossil fuels to the total energy mix during the 2010s. The findings suggest that the transition to renewables is underway in higher-income countries, but progress in lower-income countries remains slower and more uneven.

I. INTRODUCTION

Over the past 30 years, scientific research has found overwhelming evidence that the burning of fossil fuels, such as coal, oil, and natural gas, has led to significant increases in the concentration of climate-change inducing greenhouse gases within the atmosphere [1]. In 2014, the Intergovernmental Panel on Climate Change (IPCC) released a report emphasising that limiting global warming to well below 2°C is vital to prevent catastrophic damage to humans and the planet [2]. The report found that this target is only achievable through a rapid reduction in greenhouse gas emissions and a shift to renewable energy. A year later, many of these targets were ratified by countries as part of the 2015 Paris Agreement, including a global pact to aim for carbon neutrality and for developed countries to support developing countries in addressing climate change [3].

In this analysis, we will first examine how renewable energy generation has changed in recent years in three key representative regions of the world (*East Asia & Pacific*, *Europe & Central Asia*, and *Latin America & Caribbean*) to assess whether global progress is aligned with the findings discussed above. We will then explore how economic disparity affects climate progress by analysing energy usage relative to countries' GDP.

II. METHODOLOGY

Analysis of the total and relative renewable energy generation across the considered regions uses the OECD's Renewable Energy [4] and Primary Energy Supply [5] data directories. Additionally, population data from the WorldBank data repository [6] is used to provide per capita insights into energy generation.

Analysis of the relationship between GDP and electricity generation sectors uses data from the WorldBank data repository. To process and visualise the data in Python, we will use Pandas [7], NumPy [8], and Matplotlib [9].

III. ANALYSIS

A. Renewable Energy per Region

Given the global significance of climate change, both in terms of its impact on regions and the regions' contribution to it, it is crucial that our analysis adequately represents the three selected regions. Initial exploration of our datasets shows a high proportion of missing data for renewable energy generation in certain regions, which, if ignored, could skew findings. Therefore, we must carefully select a suitable time period for our analysis. We can see in Fig. 6 that the three considered regions have the least missing data across both the Renewable Energy and PES (Primary Energy Supply [10]) datasets between 1990 and 2018. During this period, the maximum proportion of records with missing data is 11.8%, whereas outside of this time period certain regions have up to 100% of data missing. The following analysis will only consider data from 1990 to 2018.

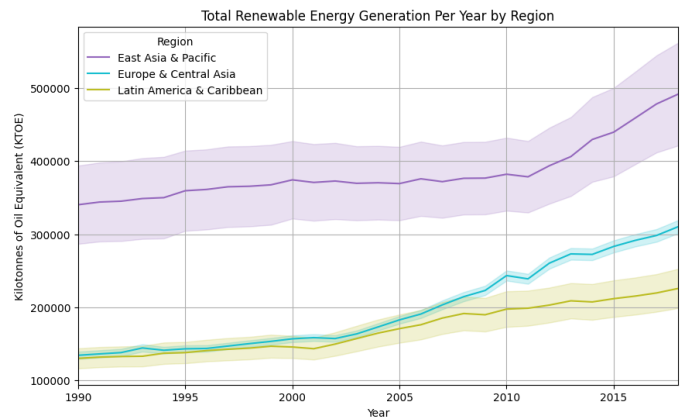


FIG. 1. Annual total renewable energy production in KTOE (Kilotonnes of Oil Equivalent) across the three selected regions (*East Asia & Pacific*, *Europe & Central Asia*, and *Latin America & Caribbean*), including the standard deviation for countries within each region, from 1990 to 2018.

During the considered time period, Fig. 1 shows that the total renewable energy produced in each of the

selected regions is steadily increasing each year. The total renewable energy production per region is highest in the East Asia & Pacific region, with its annual energy production nearly double that of Europe & Central Asia. This is most likely due to the vast land area of the East Asia & Pacific region which allows for the development of large-scale renewable energy projects, such as solar and wind farms [11].

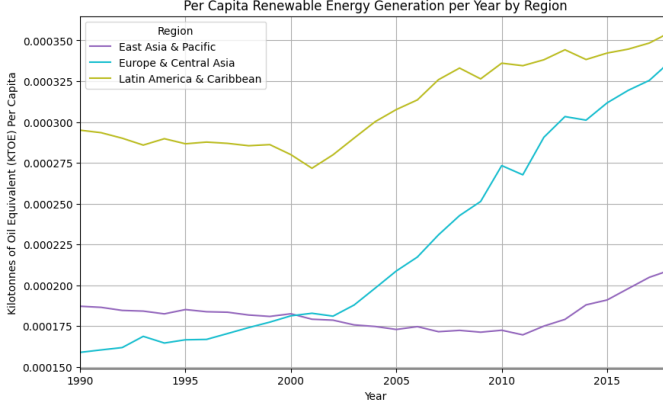


FIG. 2. Annual per capita renewable energy production in KTOE (Kilotonnes of Oil Equivalent) across the three selected regions from 1990 to 2018.

While the significant and growing renewable energy generation in East Asia & Pacific, the largest region in the world by land area, appears promising, it is important to consider these figures in the context of both population and total electricity generation. Examining Fig. 2, we notice that the renewable energy generation per capita for East Asia & Pacific is both significantly lower and growing more slowly when compared to Europe & Central Asia and Latin America & Caribbean.

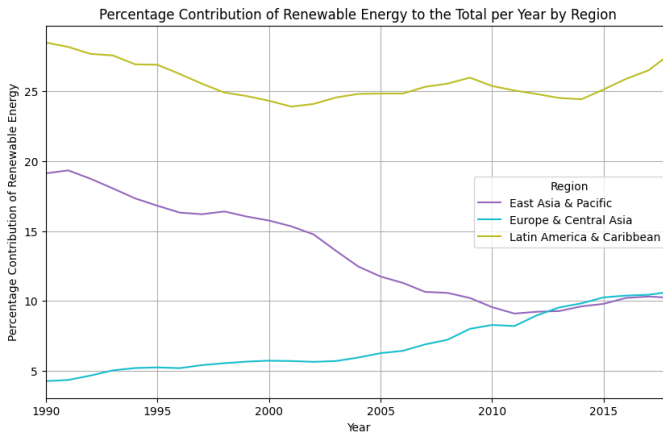


FIG. 3. Annual percentage contribution of renewable energy production to the PES (Primary Energy Supply) across the three selected regions from 1990 to 2018.

This is further evidenced by Fig. 3, which shows a downwards trajectory in the contribution of renewable energy to the total energy usage in East Asia & Pacific, falling from almost 20% in 1990 to 10% in 2010. Additionally, considering the high standard deviation for the total renewable energy generation between countries within East Asia & Pacific, as shown in Fig. 1, it is clear that while wealthier, resource-rich countries such as China, Japan, and Australia generate high amounts of renewable energy, this trend is not reflected within the lower-income, yet populous, countries of East Asia & Pacific. This economic and geographic imbalance means that despite high total renewable energy production, the region struggles to keep pace with its rapidly growing energy demand. As a result, the renewable energy share remains insufficient to drive a meaningful energy transition across the world's largest region.

The Europe & Central Asia and Latin America & Caribbean regions, on the other hand, exhibit steady growth in both total and per capita renewable energy generation, indicative of government policy to transition towards renewable energy sources [12, 13]. The relatively low standard deviation in their total renewable energy production further suggests consistent, region-wide efforts to improve renewable energy infrastructure.

Latin America & Caribbean's strong history of renewable energy development is evident in Fig. 3. In 2000, renewable energy already contributed to around 25% of their total PES, compared to 5% in Europe & Central Asia. Additionally, Fig. 2 shows that the region's per capita renewable energy generation was nearly double that of both Europe & Central Asia and East Asia & Pacific. However, despite this strong foundation, Fig. 1, Fig. 2, and Fig. 3 all show us that renewable energy growth has been slower than in Europe & Central Asia, likely due to lower average income [6] and related economic challenges. The impact of GDP on renewable energy development is explored in the next section.

B. Electricity Sector Contributions

A key indicator of net-zero progress, as outlined in the Paris Climate Agreement, is a reduced reliance on fossil fuels (notably coal, oil, and natural gas) in electricity generation. In 2000, fossil fuels accounted for almost 90% of global commercial energy [14], so transitioning away from fossil fuel dependence requires substantial economic investment in new energy infrastructure.

Fig. 4 illustrates a significant inverse relationship between economic development (GDP) and fossil fuel dependency for electricity production during the 2010s. Countries with lower GDP exhibit a wide range of fossil fuel dependence (0-100%), whereas higher-income

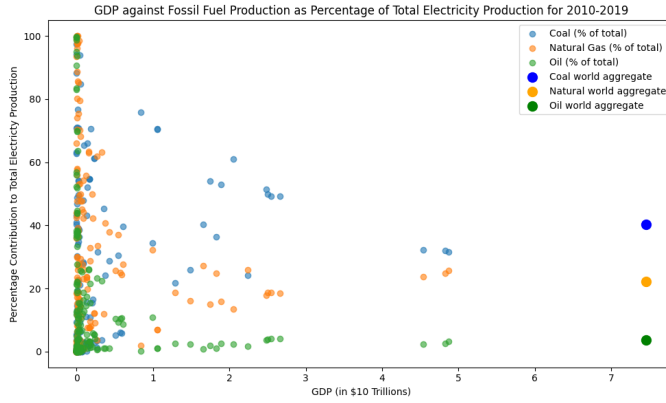


FIG. 4. Percentage of total energy production derived from fossil fuels (coal, oil, natural gas) compared to national GDP from 2010 to 2019.

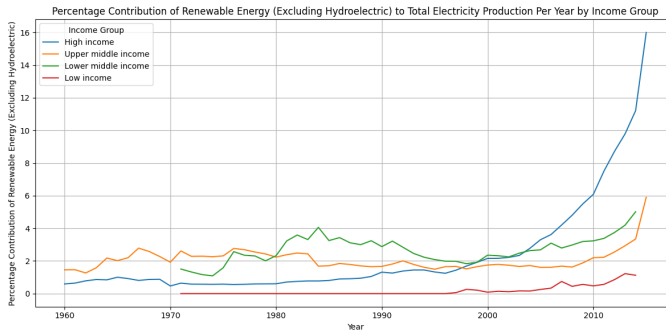


FIG. 5. Percentage of total energy derived from renewable energy (excluding hydroelectric) by country income group from 1960 to 2015.

economies show a consistent trend of reduced reliance on fossil fuels. This supports the claim that economic development plays a significant factor in a country's transition away from fossil fuels. While all fossil fuel

dependence reduces as GDP increases, there is a clear hierarchy of coal (40% world aggregate), followed closely by natural gas (22% world aggregate) and then, with a far lower contribution in economies of most sizes, oil (4% world aggregate). However, the scattered distribution of data points for similar GDP values indicates that factors beyond economic development, such as policy decisions, natural resources and historical development also influence a country's energy mix [15].

Having discovered that countries with a higher GDP tend to have less reliance on fossil fuels, we will now explore how a country's income status affects its renewable energy reliance. As shown in Fig. 5, high income countries have experienced a rapid growth since 2000, with the contribution of renewable energy (excluding hydroelectric) increasing from 2% to 16% in 2015. Although lower income group countries display less significant growth up to 2015, they appear to exhibit similar growth patterns with a 10-year lag.

IV. CONCLUSION

This study analysed renewable energy and fossil fuel prevalence across *East Asia & Pacific, Europe & Central Asia and Latin America & Caribbean*. While total renewable energy generation is increasing across all regions, per capita renewable energy generation is decreasing in East Asia & Pacific, the world's largest region by land area, highlighting the impact of population density and economic disparity. We also found that GDP and fossil fuel reliance are inversely correlated, with wealthier countries reducing their fossil fuel usage more rapidly. Furthermore, higher-income countries also demonstrated faster growth in renewable energy contributions. In conclusion, the transition from fossil fuels to renewable energy varies significantly between nations, influenced by economic factors, policy, and geography.

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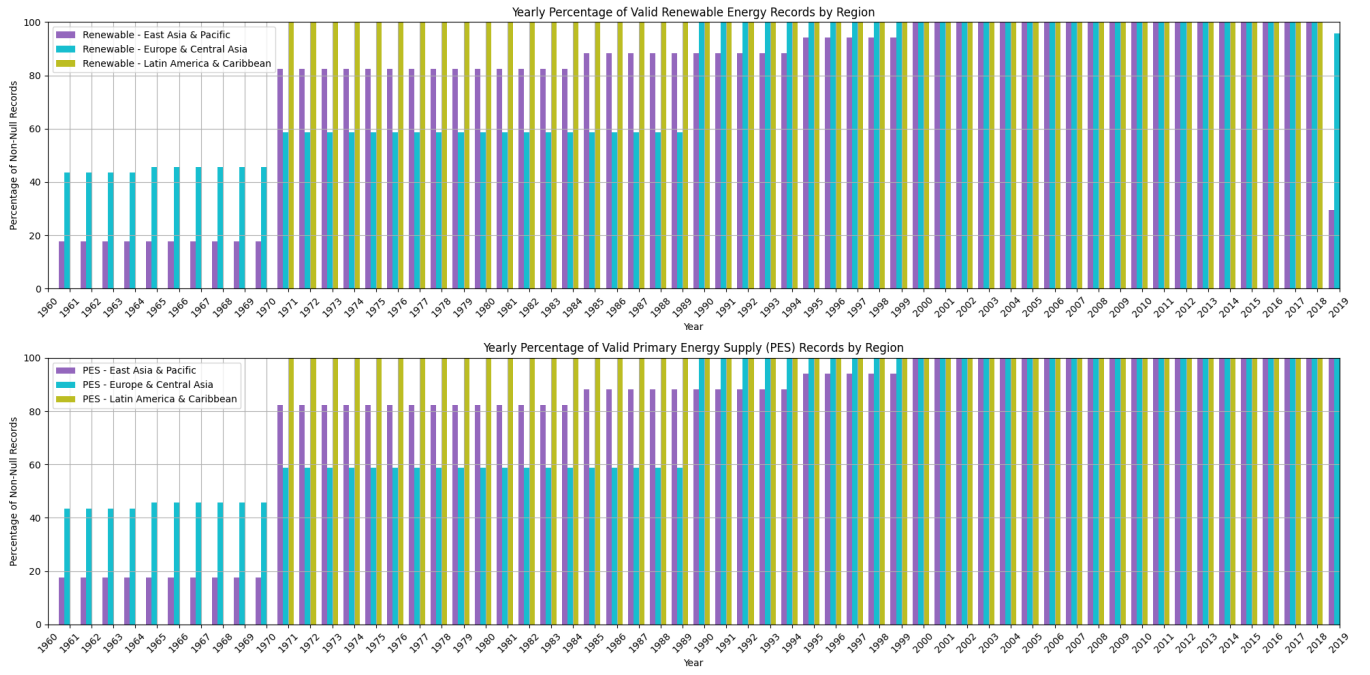


FIG. 6. Annual percentage of valid (non-null) values for Renewable Energy and PES (Primary Energy Supply) across the three selected regions for the entire dataset period of 1960 to 2019.